A Complex Event Processing Architecture for Energy and Operation Management

Industrial Experience Report

Jimi Y. C. Wen　Gu Yuan Lin　Today Sung

Minsiong Liang　Gary Tsai　Ming Whei Feng

Networks and Multimedia Institute, Institute for Information Industry, Taipei, Taiwan
{jimi, guyuan, today, minsiong, garytsai, mfeng } @nmi.iii.org.tw

ABSTRACT

In this report, we share our experience in developing a complex event processing architecture that bridges sensors networks to an energy management system used in the context of chain convenient stores. We analyze event data in real-time to generate immediate and predictive appliance/operation insights and enable instant response defined by simple business rules. For intuitive rule management, preprocessed events should be used in many cases instead of raw events collected from sensor network. We illustrate practical energy and operation management rules based on preprocessed events such as forecasted and classified events in addition to raw events.

Categories and Subject Descriptors

H.2.8 [**Database Management**]: Database Applications—*Data Mining, Prediction*; H.3.5 [**Information Storage and Retrieval**]: On-line Information Services—*Commercial services, Web-based services*; H.4.2 [**Information Systems Applications**]: Types of Systems—*Decision support*; I.5.1 [**Pattern Recognition**]: Models—*Statistical*; I.5.4 [**Pattern Recognition**]: Applications—*Signal Processing*; J.1 [**Administrative Data Processing**]: [Business]

General Terms

Algorithm, Performance, Design, Languages

1. INTRODUCTION

In this report, we share our experience in developing a complex event processing architecture that bridges sensors networks to an energy management system used in the context of chain convenient stores. Complex event processing (CEP) is the analysis of event data in real-time to generate immediate insight and enable instant response to changing conditions [6]. Sensors networks (SN) are an important type of event data in the context of an EMS and SNs weave a picture of store operation from energy consumption. Continuous streams of sensor readings can be tapped and fed into analytical algorithms in real time for predictive analytics (PA) and generate alerts for possible response actions [11]. Integrating PA into CEP architectures opens a wide range of possibilities in several proactive computing application where the response action should be triggered earlier in anticipation before the occurrence event.

We differentiate three types concepts of events in our CEP architecture and discuss their role in the context of the EMS: I) *Raw events*: frequently updated time stamped (FUTS) data energy consumption data. This first type of events deal with interfacing the database with the CEP platform and event producer such as sensors and other FUTS data sources. II).(a) *Preprocessed event-forecasted*: the likelihood of the events forecasted using mathematical methods, when the maximum energy consumption is above given thresholds, based on *Raw events*. II).(b) *Preprocessed event-classified*: a derived event using artificial intelligence methods (AI) by classifying *Raw events* into store operation events. These second type of events deal with intelligent domain-specific data analysis, where the forecasting or classification algorithm can be packaged as configurable computational plugins and used in the higher level rule language. III) *Derived event*: a higher level event defined by declarative rules on raw or previously preprocessed events. The third type of events represents the event-condition of event-condition-action (ECA) rules, which is particularly suited to event-driven monitoring and control of, energy consumption and store operation.

We illustrate the effectiveness of our CEP architecture in the context of minimizing the maximum contracted demand (MCD [10]) contracts in large-scale chain convenience stores and monitoring the store's standard operating procedure (SOP) management. For effective and intuitive rule management in the application context, derived events are often needed, i.e., writing practical applications based raw events or sensor level data, would increase the complexity in the rule management. It is better to balance the 'C' in CEP in both the event condition and the processing of data for more intuitive and powerful application development.

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2. APPLICATION OBJECTIVES

Our CEP platform is implemented and experimented with the application objective of monitoring store operation with energy-econ-awareness.

2.1 Maximum Contracted Demand Contract

A MCD contract is an agreement between the consumers and the utilities on the maximum demand load that the consumer plans to use for a given time period [10]; if the consumers use more than the agreed load, they are charged a high penalty. This type of contract is advantageous in two ways: i) Knowledge of these MCD contracts allow the utilities a better estimate on demand, therefore the utilities can plan more effective electricity generation and transmission infrastructure. ii) The consumers reduce their electricity cost if they use their electricity more effectively under the agreed maximum demand load while not necessarily decreasing their overall electricity consumption, i.e., equivalent production.

2.2 Store Operation

A practical EMS should not be a system of only minimizing energy consumption or energy cost, but it should taken into account of the context in which the EMS is to be used. In a convenience store, the context can be defined by the store's SOP, where sale performance might be more valued than pure energy related performance. We focus on the part of the SOP related to equipment and appliance operation in our EMS, e.g., switching off the refrigerator would minimize energy consumption, but this course of action could lead to a decrease of cold drink sales in the hot summer day. Thus maybe the optimal point would be increasing refrigerator temperature a degree or two which would reduce the energy consumption by a bit, but retain the same number of sales. Similarly a home EMS should taken into account comfort and convenience of the residents. Thus it is important to be able to set different rules for different contexts.

3. RELATED WORK

There are increasing interests in improving software solutions using complex event processing in many different applications [4], e.g., logistics, operation monitoring, information dissemination, threat Detection, financial applications, health monitor, smart homes and cities, gaming etc. Klien and Jerzak applied their implementation framework in the context of a sustainable manufacturing plant automation, covering the whole end-to-end energy life cycle of products starting from production and ending with daily use [5]. Guerra et al. presented, based on the Intensive Care Unit environment of the University of Utah Health Sciences Center, a system that: i) integrates in one place historical data, events, rules, and data mining models; ii) is highly customizable letting users create or change rules; and iii) identifies possible future risks by performing data mining in soft-real-time [3]. Tóth implemented a proof of concept on real world data to extend their CEP with predictive capabilities in the context of conference prediction in a given space [11]. We implement our CEP in the context of store operation monitoring with energy-econ-awareness.

Klien and Jerzak proposed a GINSENG Data Processing Framework (DPF), which is composed of a stateless publish/subscribe system, traditional event stream processing system and business rule processing engine [5]. Their DPF closes the gap between the devices and the business management systems. In order to infer high-level phenomena, Wun et al. proposed a semantic engine such that sensor data can be filtered, aggregated, correlated, and translated from many heterogeneous and dispersed sensor networks [13]. Ranganathan developed tools that allow domain experts to construct, parameterize and deploy specific flows that follow a certain pattern [9]. These 'functional' patterns help in codifying best practices in terms of event processing flows in different domains. In this paper, we focus on specific flows at the sensor-level data, i.e., forecasted events and classified events, such that domain experts can write down relevant rules to define events at management level. Our *preprocessed events* not only close the gap but the vocabulary of the devices is increased such that business rules can be applied to more applications. Our contribution includes new scope in semantics, thus more high-level phenomena can be inferred.

4. ARCHITECTURE OVERVIEW

In this section, we give a detailed description of our CEP architecture as seen in Fig. 1.

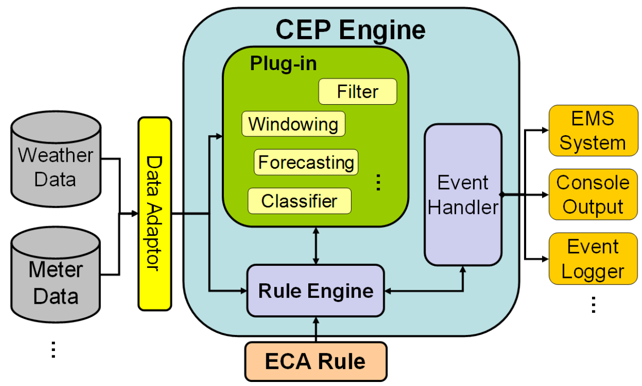

Figure 1: CEP platform for EMS system.

4.1 Raw Events

For flexible and scalable deployment of our sensor network (as can be seen in Fig. 2), we used power meters, iMeter equipped with our own developed Zigbee communication module, which is then connected to a gateway. This allows the server to control the iMeter via the internet. Each iMeter can measure 12 single-phase circuits and the voltage, current, power factor, real power, apparent power and energy can be recorded for the store and for each appliance. For the database server we use a push-pull strategy, where the database server send query to the iMeter circuits. These circuits have been monitored since 2010 July, with a total of 566 circuits available for query. Another source we use in our system is from Yahoo! Weather RSS Feed[1] which includes attributes such as, timestamp, temperature, wind direction, wind speed, humidity, visibility, air pressure, the time of sunrise and sunset, and weather condition.

4.2 Preprocessed Events

Many different processed event types are necessary for complex event processing, because it is intuitive to keep rule

[1]http://developer.yahoo.com/weather/

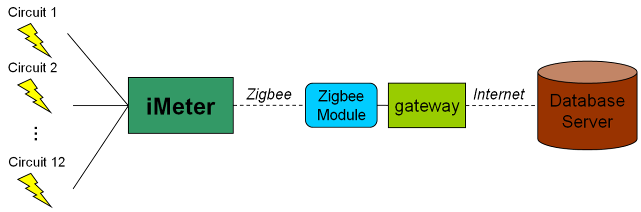

Figure 2: Sensor Network Deployment.

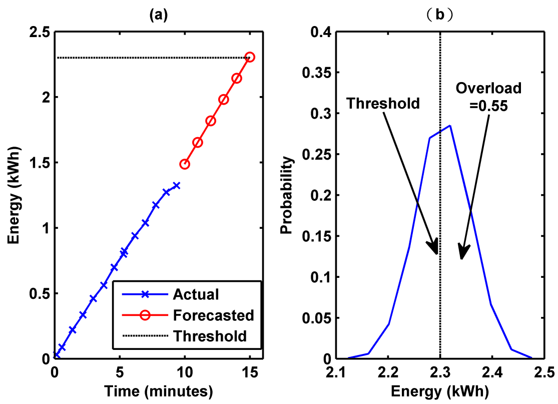

Figure 3: Load forecasting events; (a) time series forecast, (b) likelihood of event.

management simple for application users. We want system operators to deal with higher level business rules, we do not expect system operators to be writing rules to do complex computation of power signals. We discuss briefly two types of preprocessed events that were packaged as plug-ins used in our CEP in our EMS.

4.2.1 Forecasted Events

In our initial implementation, we use causal forecasting methods using time factor, i.e., a similar day of week and similar time of day approach [12]. We use 15 minute intervals, a good trade off between time resolution and sample size for consistent modelling. For each store we create a simple statistical model for each similar load profile. We then carry out numerical simulations over a large number of outcomes based on the likelihood of our load profile statistical models. Since the parameters of our application, power consumption, is dynamic by nature, we choose to implement this plug-in using the simulation approach instead of the online scoring approach [1]. The forecasted event for overload in the contract period can be attributed with a probability distribution of the load as can be seen in Fig. 3, which can be discretized into events based on the likelihood and used for higher level rule management.

4.2.2 Classified Events

We monitor each appliance, such as central air conditioner, different set of refrigerators, lighting, etc. Since the measurement is continuous, and needs to be classified into states for easier rule management. Although most events regarding appliances comprise of on-off states which can be classified using simple threshold classifiers, they can be implemented using ECA rules. We have some fault detection events for some appliances which require a domain-knowledge classifying algorithm. Preprocessed events from sensor signals, forecasted or classified can be used in the rule manager for useful energy or operation management, i.e., SOPs can easily be defined in the rule manager in terms of derived events.

4.3 Rule-based Events

For the rule manager which comprise of the rule engine and event handler, we use a RIF-like[2] syntax to illustrate some of the basic SOP events expressed in ECA rules:

- ***Raw event*** only: if sensor data *dropout*, message for system maintenance appointment.

```
Forall ?S ?M ?R (
If And ( "op:time-now" ( ?TN )
         "Sensor:Data"( ?S ?TN ?D )
         "Data:TimeStamp"( ?D ?T )
         "op:time-after" ( ?TN "5m" ?T)
         "Event:Message" ("Event:DataLoss" ?M )
         "Event:Receiver" ("Receiver:Maintenance" ?R )
Then Do( Action(?M ?R "op:Send-Message") ) )
```

- ***Preprocessed event-classified*** only: if billboard is not *on* anytime from 5pm to 5am, alert store manager to switch the billboard on.

```
Forall ?B ?M ?R (
If And ( "op:time-now" ( ?T )
         "Billboard:PowerOn"( ?B ?OOn )
         "op:time-between"( ?T "17:00" "05:00" )
         "op:text_match"( ?OOn "Off" )
         "Event:Alert" ("Event:BBPowerOff" ?M )
         "Event:Receiver" ("Receiver:Manager" ?R )
Then Do( Action(?M ?R "op:Send-Alert") ) )
```

- ***Preprocessed event-forecasted*** only: If overload probability is high, alert store manager

```
"Event:OverLoad" (?Ev) :-
Exists ?S (
And ( "op:time-now" ( ?T )
      "LoadForcast:Shop"( ?S ?T, ?LA)
      "PowerConsumption:Contract" ( ?S ?PC )
      "op:greater-than"( ?LA ?PC ) ) )
Forall ?Ev ?R (
If And ( "Event:ReadBuffer"( ?Ev "Event:OverLoad" )
         "Event:Receiver" ("Receiver:Manager" ?R )
Then Do( Action(?Ev ?R "op:Send-Alert") ) )
```

We now show some examples that incorporate combination of different layers of event to create more powerful and richly described SOP events:

- If overload probability is high and oden[3] cooker is on and outdoor temperature is above 30°C, then alert to switch off the oden cooker.

```
"Event:OdenOff" (?Ev, ?T) :-
Exists ?Ev ?O (
And ( "Event:ReadBuffer"( ?Ev "Event:OverLoad" )
      "Electricity:ODenCooker"( ?O )
      "Electricity:PowerOn"( ?O, ?OOn )
      "Environment:Temperature"( ?TM)
      "op:greater-than"( ?TM "30" )
      "op:text_match"( ?OOn "On" )
      "op:time-now" ( ?T ) ) )

Forall ?Ev ?O (
If And ( "Event:OdenOff"( ?Ev ?T )
```

[2]http://www.w3.org/TR/2010/NOTE-rif-overview-20100622/
[3]http://en.wikipedia.org/wiki/Oden

```
            "op:time-now" ( ?TN )
            "Electricity:PowerOn"( ?O, ?OOn )
            "op:text_match"( ?OOn "On" )
            "op:time-after" ( ?TN "5m" ?T)
    Then Do( Action(?O "op:power-off") ) )
```

5. DISCUSSIONS AND CONCLUSIONS

Etzion and Niblett suggested the future of CEP to be 'going from monolithic to diversified', 'going from stand-alone to embedded', 'intelligent event processing' [2]. We do not aim to build a general purpose CEP, but specific to EMS application and embedded in our EMS. We try to illustrate so called intelligent event processing by classifying raw sensor data which could cause uncertainty or inexactness, and by introducing forecasted events, our event processing is proactive rather than reactive.

Mednes et al. proposed a framework for benchmarking CEP engines [8], which include latency, throughout, accuracy, etc. Mendes et al. then evaluated micro-benchmarks on three CEP engines [7]. While in the context of high frequency trading, internet activity monitoring deal with events in seconds or milliseconds, evaluation metrics such as latency and throughput are key performance indices. Our application deals with time resolution in minutes, since sensor signal at a higher sampling rate would just include more noise and burden the computation. The events in our application comprise of human activity events, in which time resolution in minutes is more appropriate. Therefore we believe a more suitable benchmarks should include application specific performance: accuracy, electricity bills or potentially with the integration of point of sale systems, revenue analysis. These type of performances are harder to measure quantitatively and can be considered to be independent of the CEP engine. From the industry point of view, how enabling the CEP is in creating valued added solutions is the most important benchmark.

Experiments in six store locations show promising results and benefits of the proposed CEP architecture/platform for our proactive application scenario. We implemented simple data integration and developed and integrated intelligent algorithms into our application. These intelligent algorithms can be configured in the rule manager for rapid (re)development of high level energy and operation management. SOPs of the stores can then be enhanced and modified for better store KPI by monitoring event patterns with business objectives in mind.

6. ADDITIONAL AUTHORS

Additional authors: David Wu (Networks and Multimedia Institute, Institute for Information Industry, email: david.wu@nmi.iii.org.tw)

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